Functoriality as Renormalization — Langlands Lifts and the Renormalization Group

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Abstract

Langlands functoriality — the lifting of automorphic representations from one group G to another group H via an L-homomorphism ${}^LG \to {}^LH$ — is identified with renormalization group flow in the adelic quantum theory. At each place v of $\mathbb{Q}$, the local lifting $\pi_v^{(G)} \to \pi_v^{(H)}$ corresponds to an RG transformation that changes the effective anyon model: the fusion ring $R(G)_k$ flows to $R(H)_\ell$ under the renormalization step. The critical exponents of the RG flow are the special values of the associated L-function $L(s, \pi, r)$ at integer arguments. For the symmetric power liftings $\mathrm{Sym}^n : \mathrm{GL}(1) \to \mathrm{GL}(n+1)$, the RG flow connects the $\mathrm{SU}(2)_k$ anyon model at level k to effective $\mathrm{SU}(n+1)$ theories at renormalized couplings. The Langlands dual group ${}^LG$ is identified as the RG flow group — the group of admissible renormalization transformations of the adelic quantum theory. This is a conjecture paper; all claims are tagged as [speculative] or [my conjecture] as appropriate.

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1. Introduction

The Langlands program's functoriality conjecture states that for a homomorphism $\rho : {}^LG \to {}^LH$ between L-groups, there should exist a transfer of automorphic representations from $G(\mathbb{A})$ to $H(\mathbb{A})$ that preserves L-functions:

$$\Lambda(s, \pi, r \circ \rho) = \Lambda(s, \Pi, r)$$

for every finite-dimensional representation r of ${}^LH$.

In the physical framework developed by the Adelic Langlands Physics program (Phases 1—2.2), automorphic representations $|\pi\rangle$ are basis states of the adelic Hilbert space $\mathcal{H}_\mathbb{A} = \bigotimes_v' \mathcal{H}_v$, and L-functions are expectation values of the operator $\hat{L}(s)$. This paper proposes that **functoriality is renormalization group flow** in this physical setting. [my conjecture]

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2. Functoriality as RG Transformation

2.1 Local Lifting as Coupling Flow

At a fixed place v, let $\pi_v^{(G)}$ be a local automorphic representation of $G(\mathbb{Q}_v)$, corresponding under the Silent Parameter (Phase 1) to a simple object $j_v^{(G)}$ in the fusion ring $R(G)_{k_v}$. The local Langlands correspondence associates to $\pi_v^{(G)}$ a Galois parameter $\phi_v : W_{\mathbb{Q}_v} \to {}^L G$.

The L-homomorphism $\rho : {}^L G \to {}^L H$ composes with ϕ_v to give a Galois parameter into ${}^L H$, and hence (by local Langlands for H) an automorphic representation $\Pi_v^{(H)}$ of $H(\mathbb{Q}_v)$ — the **functorial lift** of $\pi_v^{(G)}$.

At the level of fusion rings, this is a homomorphism:

$$\rho_v^* : R(G)_{k_v} \longrightarrow R(H)_{\ell_v}$$

mapping the fusion ring of the "UV" theory at level k_v to the "IR" theory at renormalized level ℓ_v. The map ρ_v^* is an **RG flow step** — it reduces the number of degrees of freedom while preserving the L-function structure:

$$L_v(s, \rho_v^*(j)) = L_v(s, r \circ \phi_v)$$

[my conjecture]

2.2 Example: Symmetric Power Liftings

For $G = \mathrm{GL}(1)$ and $H = \mathrm{GL}(2)$, the symmetric square lifting $\mathrm{Sym}^2 : \mathrm{GL}(1) \to \mathrm{GL}(2)$ sends a character χ of $\mathbb{Q}_p^\times$ to a representation of $\mathrm{GL}(2, \mathbb{Q}_p)$ whose L-function is:

$$L(s, \mathrm{Sym}^2(\chi)) = L(s, \chi) \cdot L(s, \chi^2)$$

The physical interpretation: the $\mathrm{SU}(2)_k$ anyon model (describing GL(1) at conductor m) is the UV fixed point, and the $\mathrm{SU}(3)_\ell$ anyon model (describing GL(2)) is the IR fixed point reached after two RG steps — first squaring the character (doubling the anyon charge), then composing with the local transfer. [speculative]

The critical exponent ν of this RG flow is related to the L-function special value:

$$\nu = \frac{1}{\log p} \cdot \frac{d}{ds} \log L(s, \mathrm{Sym}^2(\chi)) \Big|_{s=1}$$

[speculative]

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3. The Renormalization Group Group

3.1 Adelic RG Transformations

An RG transformation on the adelic Hilbert space is an operator $\mathcal{R}$ that:

1. **Coarse-grains:** Reduces the conductor $m_v \to m_v - 1$ at each finite place v (eliminating the highest-resolution anyon types).
2. **Renormalizes the coupling:** Adjusts the level $k_v \to \ell_v$ according to the lifting map.
3. **Preserves L-function structure:** $\Lambda(s, \mathcal{R}|\pi\rangle) = \Lambda(s, \pi)$ up to a finite Euler factor.

The set of all such transformations forms a **group** — the RG group $\mathcal{G}_{\mathrm{RG}}$.

3.2 Langlands Dual Group as RG Group

Conjecture 1 (RG-Langlands Duality). The group of admissible adelic RG transformations $\mathcal{G}_{\mathrm{RG}}$ is isomorphic to the Langlands dual group ${}^L G$ of the adelic quantum theory.

Reasoning sketch. The L-group ${}^L G = \hat{G} \rtimes W_{\mathbb{Q}}$ has two components: the complex dual group $\hat{G}$ (e.g., $\mathrm{SL}(n, \mathbb{C})$ for $G = \mathrm{SU}(n)$) and the Weil group action. The $\hat{G}$ component corresponds to the continuous part of the RG flow — the family of coupling constants parametrized by the dual group. The Weil group action corresponds to the discrete scaling transformations (changes of conductor). [my conjecture]

For GL(1), the L-group is simply $\mathrm{GL}(1, \mathbb{C}) \cong \mathbb{C}^\times$, which is exactly the multiplicative group of complex numbers — the scaling group of the β-function in ordinary QFT.

3.3 β-Function of the Adelic Theory

The adelic β-function is defined as the infinitesimal generator of the RG flow:

$$\beta(g_v) = -\frac{dg_v}{d\log\mu}$$

where g_v is the coupling at place v and μ is the energy scale. The zeros of $\beta(g)$ are the fixed points — corresponding to the **automorphic representations with trivial epsilon factors** (the unramified principal series).

The special values of L-functions provide the critical exponents:

$$\eta = \lim_{s \to 1} (s-1) \cdot \frac{d}{ds} \log \Lambda(s, \pi)$$

which is the anomalous dimension of the automorphic field. [speculative]

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4. Physical Examples

4.1 Central Charge Flow

For anyon models, the central charge c (the chiral central charge of the edge CFT) flows under RG according to Zamolodchikov's c-theorem: $c_{\mathrm{UV}} \geq c_{\mathrm{IR}}$.

In the Langlands framework, the central charge at each place is:

$$c_v = \log|\varepsilon_v|^2$$

where ε_v is the local epsilon factor (Phase 2.2). The RG flow p-adic place $\to$ ∞-place sends $c_p^{\mathrm{UV}} \to c_\infty^{\mathrm{IR}}$, with the monotonic decrease $\Delta c = \log|\Lambda(1,\pi)|$ given by the special value of the L-function. [speculative]

4.2 Symmetric Power Cascade

For the symmetric power lifting chain:

$$\mathrm{GL}(1) \xrightarrow{\mathrm{Sym}^2} \mathrm{GL}(2) \xrightarrow{\mathrm{Sym}^3} \mathrm{GL}(3) \xrightarrow{\mathrm{Sym}^4} \cdots$$

each step increases the rank of the target group by 1. The corresponding RG cascade connects $\mathrm{SU}(2)_k \to \mathrm{SU}(3)_\ell \to \mathrm{SU}(4)_m \to \cdots$, with the levels flowing according to:

$$\ell = \mathrm{Sym}^2(k), \quad m = \mathrm{Sym}^3(\ell), \quad \ldots$$

At each step, the anyon content becomes richer, and the physical theory flows from simpler (fewer anyon types) to more complex (more anyon types) — the opposite of the usual coarse-graining RG direction. This is **inverse RG flow**, characteristic of topological orders connected by anyon condensation. [speculative]

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5. Falsifiable Predictions

1. **[P1] Functoriality $\leftrightarrow$ RG flow.** For any L-homomorphism $\rho : {}^L G \to {}^L H$, there exists an RG transformation $\mathcal{R}_\rho$ on the adelic Hilbert space such that $\mathcal{R}_\rho|\pi^{(G)}\rangle = |\Pi^{(H)}\rangle$ for all automorphic representations π of G. [my conjecture]
   - **Falsification:** Find a functorial lift that cannot be realized as an RG flow on the fusion ring.
   - **[CHECK: 2028-12]** Systematic enumeration of low-rank liftings.
2. **[P2] Critical exponents from L-functions.** The critical exponents of the RG flow $\mathcal{R}_\rho$ are the logarithmic derivatives of the associated L-function at integer arguments. [speculative]
   - **Falsification:** Compute critical exponents from first-principles RG and compare to L-function special values.
   - **[CHECK: 2029-06]** Numerical simulation for Sym^2 at small primes.
3. **[P3] c-theorem from Langlands.** The central charge $c(\pi) = \log|\Lambda(1,\pi)|$ is monotone decreasing under any Langlands functorial lifting. [my conjecture]
   - **Falsification:** Find a lifting where c increases.
   - **[CHECK: 2028-06]** Verlinde formula computation for $\mathrm{SU}(2)_k \to \mathrm{SU}(n)_\ell$ at small k.

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6. Conclusion

Phase 2.3 completes the global Langlands construction by identifying **functoriality as renormalization group flow**. The L-group ${}^{L}G$ is the RG group of the adelic quantum theory, and the critical exponents of the flow are the special values of L-functions. Together with Phases 2.1 (Hecke operators as quantum gates) and 2.2 (L-function as physical observable), Phase 2 establishes the global framework connecting automorphic representations to physical quantum systems across all completions of $\mathbb{Q}$.

Phase 3 will construct the full adelic gauge theory — the field theory on the adele ring that unifies $N = 4$ SYM S-duality at the ∞-place with anyon braiding at finite primes.

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